

human

Where are houses in Exeter that were built after 2000?

host_agent

Calling tool `send_message` with args:

```
{'target': 'planning_agent', 'task_description': 'User wants to find houses in Exeter that were built after 2000. Provide the general steps to solve this geospatial query.'}
```

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is houses, and the condition is that they must have been built after 2000.

output steps: ["Find Exeter 1 area", "Search for houses in Exeter as many entries as possible", "Apply condition for houses built after 2000"]

host_agent

Tool `send_message` returned result: The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is houses, and the condition is that they must have been built after 2000.

output steps: ["Find Exeter 1 area", "Search for houses in Exeter as many entries as possible", "Apply condition for houses built after 2000"]

Calling tool `send_message` with args:

```
{'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}
```

named_area_agent

Calling tool `call_osngd` with args:

```
{'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_polygon'}
```

Tool `call_os_ngd` returned result: - Search results have been found. - Artifact generated is: `exeter_polygon` - Description: A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter. - Count of records fetched: 98

Calling tool `send_message` with args:

```
{'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_polygon' to find the specific area polygon for Exeter. Save the filtered re
```

data_analysis_agent

Calling tool `generate_metadata_for_artifacts` with args:

```
{'artifact_names': ['exeter_polygon']}
```

Tool `generate_metadata_for_artifacts` returned result: (metadata for 'exeter_polygon')

Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the name or description contains 'Exeter' (case-insensitive)
    mask = (
        gdf['name1_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name2_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name3_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name4_text'].str.contains('Exeter', case=False, na=False) |
        gdf['description'].str.contains('Exeter', case=False, na=False)
    )
    filtered = gdf[mask]
    summary = f"Filtered polygons for Exeter. Number of matching polygons: {len(filtered)}."
    return [summary, 'filtered_exeter_polygon', 'A GeoDataFrame containing only the polygons specifically related to Exeter, filtered by name and description c
```

Tool `code_executor` returned result: - Filtered polygons for Exeter. Number of matching polygons: 1. - Artifact generated: `filtered_exeter_polygon`

Filtered polygons for Exeter. Number of matching polygons: 1. An artifact has been generated with the name 'filtered_exeter_polygon' and contains a GeoDataFrame with only the polygons specifically related to Exeter, filtered by name and description columns.

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_polygon'. This artifact contains only the polygons specifically related to Exeter.

host_agent

Calling tool `send_message` with args:

```
{'target': 'buildings', 'task_description': "Search for houses in the area defined by 'filtered_exeter_polygon' that were built after 2000."}
```

buildings_agent

Calling tool `call_os_ngd` with args:

```
{'filters': None, 'bbox': 'filtered_exeter_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_buildings_after_2000'}
```

Tool `call_os_ngd` returned result: - Multiple search results have been found in multiple datasets. A summary of each is provided. - Artifacts generated are: ['buildings_exeter_buildings_after_2000', 'buildingline_exeter_buildings_after_2000', 'buildingpart_exeter_buildings_after_2000'] - Descriptions: ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for houses in the area defined by 'filtered_exeter_polygon' that were built after 2000. using filters None.", ...] - Counts of records fetched: [69169, 69342, 85480]

Calling tool `send_message` with args:

```
{'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_exeter_buildings_after_2000' to find houses, defined as buildings with b"
```

data_analysis_agent

Calling tool `generate_metadata_for_artifacts` with args:

```
{'artifact_names': ['buildings_exeter_buildings_after_2000']}
```

Tool `generate_metadata_for_artifacts` returned result: (metadata for 'buildings_exeter_buildings_after_2000')

Tool Code Executor

```
def filter_houses_exeter_after_2000(data: list):
    gdf = data[0]
    # Filter for houses: residential > 0, total = 1, and constructed after 2000
    filtered = gdf[(gdf['buildinguse_addresscount_residential'] > 0) &
                  (gdf['buildinguse_addresscount_total'] == 1) &
                  (gdf['buildingage_year'] > 2000)]
    summary = f"Filtered {len(filtered)} houses in Exeter built after 2000."
    artifact_name = 'filtered_houses_exeter_after_2000'
    artifact_description = 'GeoDataFrame of houses in Exeter (residential, single address, built after 2000).'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result: - Filtered 5644 houses in Exeter built after 2000. - Artifact generated: filtered_houses_exeter_after_2000

Filtered 5,644 houses in Exeter that are residential, have a single address, and were built after 2000. The results are saved as 'filtered_houses_exeter_after_2000' and contain detailed information for these properties in a GeoDataFrame.

buildings_agent

I found 5,644 houses in the 'filtered_exeter_polygon' area that are residential, have a single address, and were built after 2000. The results are saved as 'filtered_houses_exeter_after_2000'.

host_agent

Calling tool send_message with args:

```
{'target': 'plotting_agent', 'task_description': "Plot the houses from 'filtered_houses_exeter_after_2000' within the area of 'filtered_exeter_polygon'."}
```

plotting_agent

Calling tool generate_metadata_for_artifacts with args:

```
{'artifact_names': ['filtered_houses_exeter_after_2000', 'filtered_exeter_polygon']}
```

Tool generate_metadata_for_artifacts returned result: (metadata for 'filtered_houses_exeter_after_2000', 'filtered_exeter_polygon')

Tool Code Executor

```

def plot_houses_within_exeter_polygon(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    houses = data[0].copy()
    exeter_poly = data[1].copy()

    # Ensure CRS is the same and set to EPSG:4326 for folium
    if houses.crs != 'EPSG:4326':
        houses = houses.to_crs('EPSG:4326')
    if exeter_poly.crs != 'EPSG:4326':
        exeter_poly = exeter_poly.to_crs('EPSG:4326')

    # Get the Exeter polygon (assuming only one row)
    exeter_geom = exeter_poly.iloc[0].geometry

    # Filter houses within the Exeter polygon
    houses_within = houses[houses.within(exeter_geom)]

    # Create folium map centered on Exeter polygon centroid
    centroid = exeter_geom.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Add Exeter polygon to map
    folium.GeoJson(mapping(exeter_geom), name='Exeter Area', style_function=lambda x: {'fillColor': 'blue', 'color': 'blue', 'weight': 2, 'fillOpacity': 0.1}).

    # Add houses to map
    for _, row in houses_within.iterrows():
        folium.GeoJson(mapping(row.geometry),
                        name=row.get('description', 'House'),
                        style_function=lambda x: {'fillColor': 'red', 'color': 'red', 'weight': 1, 'fillOpacity': 0.5}).add_to(m)

    # Save map
    map_filename = 'houses_within_exeter_polygon_map.html'
    m.save(map_filename)

    # Prepare summary
    summary_rows = houses_within[['osid', 'description', 'buildingage_year', 'geometry_area_m2']].head(5)
    summary = f"Plotted {len(houses_within)} houses built after 2000 within the Exeter polygon. First 5 entries:\n{summary_rows.to_string(index=False)}"

    return [summary, 'houses_within_exeter_polygon_map', 'Folium map showing houses built after 2000 within the Exeter polygon area. Houses are shown in red, E

```

Tool `code_executor` returned result: - Plotted 5641 houses built after 2000 within the Exeter polygon. First 5 entries: - a578be0a-7fff-4fbb-8ace-2cf755bbb53c: End-Of-Terrace House, 2021, 51.92 m² - c86c6a23-5ced-4dae-be17-94f33e02c66d: Mid-Terrace House, 2021, 51.91 m² - 5cc5cbef-f15d-4110-9493-25be6ea12946: Mid-Terrace House, 2021, 51.91 m² - 93d80cf9-5e7b-4c78-980f-c0acc04ed8a4: End-Of-Terrace

House, 2021, 51.91 m² - a363510c-79e9-4fbd-8b2b-cd971a4dfcbe: Semi-Detached House, 2021, 50.21 m² - Artifact name: houses_within_exeter_polygon_map - Description: Folium map showing houses built after 2000 within the Exeter polygon area. Houses are shown in red, Exeter boundary in blue. - File: houses_within_exeter_polygon_map.html

host_agent

I have found and mapped 5,641 houses in Exeter that were built after the year 2000. These houses are all residential, have a single address, and are located within the official Exeter area boundary.

- The map shows these houses in red, with the Exeter boundary in blue.
- Example house types include End-Of-Terrace, Mid-Terrace, and Semi-Detached, with construction years like 2021 and typical sizes around 50–52 m².

If you would like to view or download the map, let me know! If you need more details or a list of these houses, I can provide that as well.